

EKLAKH DEWAN

ML Engineer — Machine Learning — NLP — Retrieval & Evaluation

eklakh.inplace@gmail.com — github.com/eklakhdevan — linkedin.com/in/eklakhdevan — Portfolio

SUMMARY

ML Engineer focused on building and evaluating machine learning and NLP systems from unstructured data. Experienced in feature engineering, semantic embeddings, text classification, recommendation pipelines, retrieval evaluation, model validation, and reproducible Python-based ML workflows.

EDUCATION

KPRIET

B.Tech. in Artificial Intelligence & Data Science

2027

CGPA: 8.55

EXPERIENCE

Artificial Intelligence Intern

Flowrage Technology

Virtual Internship — 4 Weeks

- Integrated the **Google Gemini API** into AI-driven application workflows for resume analysis and job recommendation use cases.
- Built NLP processing pipelines to clean, transform, and represent unstructured resume and job-description text for downstream machine learning tasks.
- Applied **TF-IDF** and dense semantic embeddings to model lexical and semantic relationships between candidate profiles and job requirements.
- Developed matching and ranking logic connecting feature extraction, similarity computation, and recommendation workflows.

SELECTED MACHINE LEARNING PROJECTS

AI-Powered Job Recommendations

[GitHub](#)

Python, scikit-learn, SentenceTransformers, Streamlit

- Built a recommendation pipeline combining **TF-IDF** and dense semantic embeddings to rank job descriptions against unstructured resume content.
- Applied text preprocessing, feature representation, cosine similarity, and semantic matching to generate interpretable job recommendations.
- Developed an interactive Streamlit dashboard with skill-gap analysis to expose recommendation signals to end users.

Enterprise RAG Evaluation Pipeline

[GitHub](#)

Python, FAISS, BM25, SentenceTransformers, Retrieval Evaluation

- Developed a retrieval evaluation workflow comparing dense FAISS retrieval, sparse BM25 retrieval, hybrid retrieval, and reranked results.
- Evaluated retrieval quality using **Recall@K**, **Mean Reciprocal Rank (MRR)**, and **nDCG** across retrieval pipeline stages.
- Integrated SentenceTransformers embeddings and cross-encoder reranking to improve semantic relevance of retrieved evidence.

Insurance Claims ML Automation

[GitHub](#)

Python, XGBoost, scikit-learn, Pandas, Classification

- Designed an end-to-end machine learning workflow covering data ingestion, preprocessing, feature engineering, classification, and evaluation for insurance claims.
- Applied class-imbalance handling and predictive modeling techniques to distinguish valid claims from anomalous or potentially fraudulent cases.
- Structured the pipeline around reproducible preprocessing, model training, and precision/recall-based evaluation.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript

Machine Learning: scikit-learn, XGBoost, Classification, Feature Engineering, Model Evaluation, Predictive Modeling

NLP: TF-IDF, SentenceTransformers, Embeddings, Semantic Similarity, Text Classification, NLP Preprocessing

Retrieval & Evaluation: FAISS, BM25, Hybrid Search, RRF, Cross-Encoder Reranking, Recall@K, MRR, nDCG

Data: Pandas, NumPy, SQL, PostgreSQL, Data Cleaning, Exploratory Data Analysis

Engineering: FastAPI, Streamlit, Docker, GitHub Actions, Reproducible Pipelines

ENGINEERING FOCUS

Feature engineering — Model evaluation — NLP pipelines — Semantic retrieval — Recommendation systems — Reproducibility — Data preprocessing — Experiment-driven development